# A Priori Error Analysis

The $L^2$ result and numerical verification with linear elasticity

# Lecture outline

1. State the predicted convergence rates and the regularity assumptions behind them.
2. Trace the main steps that give the improved $L^2$ estimate.
3. Verify the predicted rates for a smooth elasticity problem in a Jupyter notebook.
4. Examine what changes when an elastic body has a reentrant corner.

The emphasis today is on interpreting and testing the error estimates. The duality argument is included so that the additional $L^2$ order is not used as an unexplained rule.

# Review: diffusion problem and variational form

With homogeneous Dirichlet data, the strong problem is

$$\begin{aligned} -\nabla \cdot (k \nabla u) &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \Gamma_D, \\ -k \nabla u \cdot \boldsymbol{n} &= q \quad \text{on } \Gamma_N. \end{aligned}$$

The test and trial space is

$$\mathcal{V}_0 = H^1_{0,\Gamma_D}(\Omega).$$

Define the bilinear and linear forms

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}\boldsymbol{x},$$

$$\ell(v) = \int_\Omega f v \, \mathrm{d}\boldsymbol{x} - \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The variational problem is

$$\boxed{\text{find } u \in \mathcal{V}_0 : \quad a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0.}$$

# Review: finite element problem and stability estimates

Let $\mathcal{V}_{h,0} \subset \mathcal{V}_0$ be the continuous Lagrange finite element space of polynomial degree $p \geq 1$. The finite element problem is

$$\boxed{\text{find } u_h \in \mathcal{V}_{h,0}: \quad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.}$$

If $0 < k_0 \leq k(\boldsymbol{x}) \leq k_1$, then

$$|a(w, v)| \leq M \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}, \qquad M = k_1,$$

$$a(v, v) \geq \alpha \|v\|^2_{H^1(\Omega)}, \qquad \alpha = \frac{k_0}{1 + C_P^2}.$$

Here $M$ is the continuity constant and $\alpha$ is the coercivity constant. Since $a$ is symmetric,

$$\|v\|_a = \sqrt{a(v, v)}$$

defines the energy norm.

# Review: orthogonality, Céa's lemma, and interpolation

Let $e = u - u_h$. For any $v_h \in \mathcal{V}_{h,0}$,

$$a(e, v_h) = a(u, v_h) - a(u_h, v_h) = \ell(v_h) - \ell(v_h) = 0.$$

This is Galerkin orthogonality. Céa's lemma gives

$$\|e\|_{H^1(\Omega)} \leq \frac{M}{\alpha} \inf_{w_h \in \mathcal{V}_{h,0}} \|u - w_h\|_{H^1(\Omega)}.$$

For degree-$p$ interpolation on a shape-regular mesh, if $u \in H^s(\Omega)$ and $2 \leq s \leq p+1$,

$$\|u - \mathcal{I}_h u\|_{L^2(\Omega)} \leq C_I h^s |u|_{H^s(\Omega)},$$

$$|u - \mathcal{I}_h u|_{H^1(\Omega)} \leq C_I h^{s-1} |u|_{H^s(\Omega)}.$$

# Review: a priori estimates obtained previously

Combining best approximation with the interpolation estimate gives

$$\|e\|_a \leq C_I \sqrt{k_1}\, h^{s-1} \, |u|_{H^s(\Omega)} .$$

The coefficient bounds give the $H^1$-seminorm estimate

$$\boxed{|e|_{H^1(\Omega)} \leq C_I \sqrt{\frac{k_1}{k_0}}\, h^{s-1} \, |u|_{H^s(\Omega)} .}$$

Using Poincaré's inequality gives

$$\|e\|_{H^1(\Omega)} \leq C_I \sqrt{\frac{k_1(1+C_P^2)}{k_0}}\, h^{s-1} \, |u|_{H^s(\Omega)} .$$

Therefore, the direct $L^2$ estimate is only

$$\|e\|_{L^2(\Omega)} \leq \|e\|_{H^1(\Omega)} \leq C h^{s-1} \, |u|_{H^s(\Omega)} .$$

# What rates should we expect?

Polynomial degree and solution regularity both limit the answer.

# Predicted rates for a smooth solution

Assume $u \in H^{p+1}(\Omega)$. The $L^2$ column also requires the dual-regularity assumption introduced next.

| degree $p$ | energy and $H^1$ error | $L^2$ error |
|---|---|---|
| 1 | $O(h)$ | $O(h^2)$ |
| 2 | $O(h^2)$ | $O(h^3)$ |
| 3 | $O(h^3)$ | $O(h^4)$ |

$E(h) = O(h^r)$ means $E(h) \leq Ch^r$, where $C$ is independent of $h$. The exponent $r$ is the convergence rate in the stated norm.

# Why can the $L^2$ error converge faster?

A dual problem supplies one additional factor of $h$.

# The dual problem associated with the error

Let $e = u - u_h$. Using the same diffusion bilinear form

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}\boldsymbol{x},$$

the **dual problem** is to find $z \in \mathcal{V}_0$ such that

$$a(v, z) = (e, v)_{L^2(\Omega)} \qquad \forall v \in \mathcal{V}_0.$$

Because $a$ is symmetric, this is the variational form of

$$\begin{aligned} -\nabla \cdot (k \nabla z) &= e \quad \text{in } \Omega, \\ z &= 0 \quad \text{on } \Gamma_D, \\ k \nabla z \cdot \boldsymbol{n} &= 0 \quad \text{on } \Gamma_N. \end{aligned}$$

We assume that $z \in H^2(\Omega)$ and

$$|z|_{H^2(\Omega)} \leq C_{\mathrm{reg}} \, \|e\|_{L^2(\Omega)} .$$

# The error identity

Choose $v = e$ in the dual problem:

$$\|e\|^2_{L^2(\Omega)} = a(e, z).$$

The $H^2$ regularity of $z$ ensures that its interpolant is defined, and $\mathcal{I}_h z \in \mathcal{V}_{h,0}$. Galerkin orthogonality states

$$a(e, v_h) = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Use the particular test function $v_h = \mathcal{I}_h z$:

$$a(e, \mathcal{I}_h z) = 0.$$

By bilinearity in the second argument,

$$\boxed{\|e\|^2_{L^2(\Omega)} = a(e, z) - a(e, \mathcal{I}_h z) = a(e, z - \mathcal{I}_h z).}$$

The right-hand side now contains the interpolation error of the dual solution.

# Obtain the additional factor of $h$

For the diffusion form, the coefficient bound and Cauchy–Schwarz inequality give

$$|a(w, v)| \leq k_1 \, |w|_{H^1(\Omega)} \, |v|_{H^1(\Omega)} \, .$$

Use this continuity estimate, interpolation with $s = 2$, and dual regularity:

$$\begin{aligned} \|e\|^2_{L^2(\Omega)} &= a(e, z - \mathcal{I}_h z) \\ &\leq k_1 \, |e|_{H^1(\Omega)} \, |z - \mathcal{I}_h z|_{H^1(\Omega)} \\ &\leq k_1 C_I h \, |e|_{H^1(\Omega)} \, |z|_{H^2(\Omega)} \\ &\leq k_1 C_I C_{\mathrm{reg}} h \, |e|_{H^1(\Omega)} \, \|e\|_{L^2(\Omega)} \, . \end{aligned}$$

# Combine with the $H^1$ estimate

If $\|e\|_{L^2(\Omega)} \neq 0$, cancel one error factor to obtain

$$\|e\|_{L^2(\Omega)} \leq k_1 C_I C_{\mathrm{reg}} h \, |e|_{H^1(\Omega)} .$$

The recalled $H^1$-seminorm estimate is

$$|e|_{H^1(\Omega)} \leq C_I \sqrt{\frac{k_1}{k_0}} \, h^{s-1} \, |u|_{H^s(\Omega)} .$$

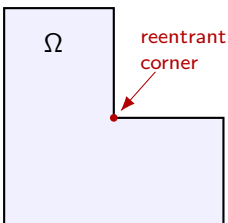
Combining the two estimates gives

$$\|e\|_{L^2(\Omega)} \leq C_I^2 C_{\mathrm{reg}} k_1 \sqrt{\frac{k_1}{k_0}} \, h^s \, |u|_{H^s(\Omega)} .$$

For degree-$p$ elements, provided the exact solution belongs to $H^{p+1}(\Omega)$ and the dual problem is $H^2$-regular, choose $s = p + 1$:

$$\boxed{\|u - u_h\|_{L^2(\Omega)} \leq C h^{p+1} \, |u|_{H^{p+1}(\Omega)} .}$$

# A corner can limit the available regularity




A reentrant corner can produce a solution that is less regular even when the data are smooth.

If the exact solution does not belong to $H^2(\Omega)$, the usual first-order $H^1$ estimate for linear elements does not follow from the theorem.

The dual solution may also fail to belong to $H^2(\Omega)$. In that case the duality argument does not supply a full additional power of $h$ in the $L^2$ estimate. Increasing $p$ does not restore missing regularity.

# Numerical verification

Test the predicted rates for a smooth elasticity problem.

# Observed convergence rate

If $E(h) \approx Ch^r$, two consecutive meshes give

$$\boxed{r_{\mathrm{obs}} = \frac{\log(E(h_1)/E(h_2))}{\log(h_1/h_2)}.}$$

For uniform refinement with $h_2 = h_1/2$,

$$r_{\mathrm{obs}} = \log_2\left(\frac{E(h)}{E(h/2)}\right).$$

Therefore:

| error reduction after halving $h$ | 2 | 4 | 8 |
|---|---|---|---|
| $r_{\mathrm{obs}}$ | 1 | 2 | 3 |

We look for stabilization of $r_{\mathrm{obs}}$ over several refinements, not agreement from a single pair of meshes.

# What a verification study requires

To compare a theorem with a computation, we need:

1. an exact solution with the regularity assumed by the theorem;
2. data chosen so that this exact solution satisfies the boundary-value problem;
3. a mesh sequence that preserves element quality;
4. the error evaluated in each norm for which a rate is predicted;
5. sufficiently accurate quadrature for the error integrals; and
6. enough refinements to reach the asymptotic range.

An attractive solution plot is not a convergence test. Nodal errors alone are also insufficient because the error between nodes may be nonzero.

# Linear elasticity problem used in the notebook

For plane strain, with $\lambda = \mu = 1$,

$$\varepsilon(\boldsymbol{u}) = \tfrac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right), \qquad \boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu\varepsilon(\boldsymbol{u}) + \lambda \operatorname{tr}(\varepsilon(\boldsymbol{u}))\boldsymbol{I}.$$

The finite element problem is

$$\int_\Omega \boldsymbol{\sigma}(\boldsymbol{u}_h) : \varepsilon(\boldsymbol{v}_h)\,\mathrm{d}\boldsymbol{x} = \int_\Omega \boldsymbol{b}\cdot\boldsymbol{v}_h\,\mathrm{d}\boldsymbol{x} \qquad \forall \boldsymbol{v}_h \in \mathcal{V}_{h,0}.$$

For a smooth exact displacement and $P_1$ triangles, the predictions are

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{L^2(\Omega;\mathbb{R}^2)} = O(h^2),$$

$$|\boldsymbol{u} - \boldsymbol{u}_h|_{H^1(\Omega;\mathbb{R}^2)} = O(h), \qquad \|\boldsymbol{u} - \boldsymbol{u}_h\|_{a_{\mathrm{el}}} = O(h).$$

# Manufactured smooth displacement

Let $\Omega = (0,1)^2$ and prescribe $\boldsymbol{u} = \mathbf{0}$ on $\partial\Omega$. Choose

$$\boldsymbol{u}_\star(x,y) = \begin{bmatrix} \sin(\pi x)\sin(\pi y) \\ \sin(2\pi x)\sin(\pi y) \end{bmatrix}.$$

The body force is then defined by

$$\boldsymbol{b} = -\operatorname{div}\boldsymbol{\sigma}(\boldsymbol{u}_\star).$$

This construction guarantees that $\boldsymbol{u}_\star$ solves the strong problem. It is smooth and is not contained in the $P_1$ finite element space.

We can therefore compute the $L^2$, $H^1$-seminorm, and energy errors directly and compare their observed rates with the predicted rates.

# Notebook workflow

For each subdivision count $n = 4, 8, 16, 32$:

1. construct a triangular mesh of the unit square;
2. define the vector-valued $P_1$ space;
3. form $\boldsymbol{b} = -\operatorname{div} \boldsymbol{\sigma}(\boldsymbol{u}_\star)$;
4. assemble and solve the elasticity problem;
5. integrate the three error norms; and
6. compute the observed rates between successive meshes.

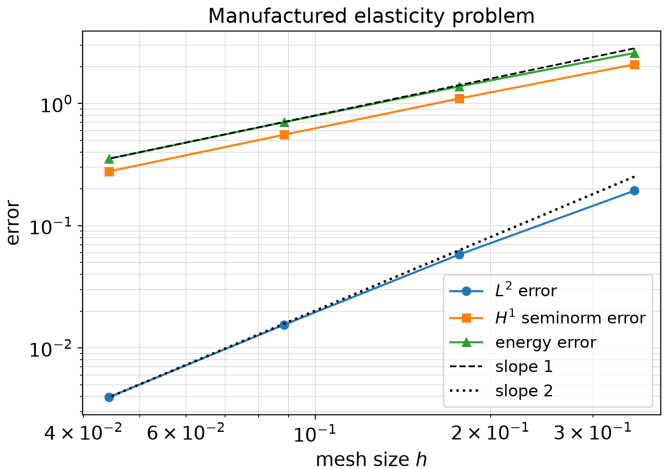
The notebook follows this sequence explicitly. Plotting helpers are kept separate so that the finite element algorithm remains visible during the demonstration.

# Computed errors and observed rates

| $n$ | $L^2$ error | $H^1$ error | energy error | $L^2$ rate | $H^1$ rate | energy rate |
|---|---|---|---|---|---|---|
| 4 | $1.931\times10^{-1}$ | 2.088 | 2.588 | – | – | – |
| 8 | $5.798\times10^{-2}$ | 1.097 | 1.378 | 1.74 | 0.93 | 0.91 |
| 16 | $1.545\times10^{-2}$ | $5.543\times10^{-1}$ | $7.020\times10^{-1}$ | 1.91 | 0.98 | 0.97 |
| 32 | $3.933\times10^{-3}$ | $2.778\times10^{-1}$ | $3.527\times10^{-1}$ | 1.97 | 1.00 | 0.99 |

The rates approach 2 in $L^2$ and 1 in both derivative-based norms. The coarsest pair is not yet fully in the asymptotic range.

# The log–log plot shows the same rates




A curve parallel to a reference line of slope $r$ behaves like $h^r$.

The $L^2$ curve approaches slope 2.

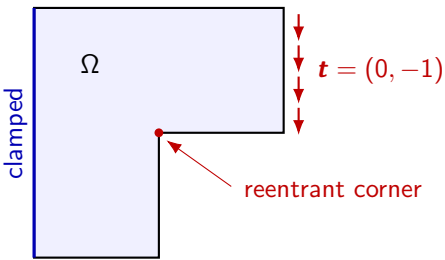
The $H^1$-seminorm and energy-error curves approach slope 1.

The table is still needed: a plot alone makes precise rate differences difficult to judge.

# An elastic body with a reentrant corner

Separate numerical verification from a study without a known exact solution.

# Geometry and boundary conditions

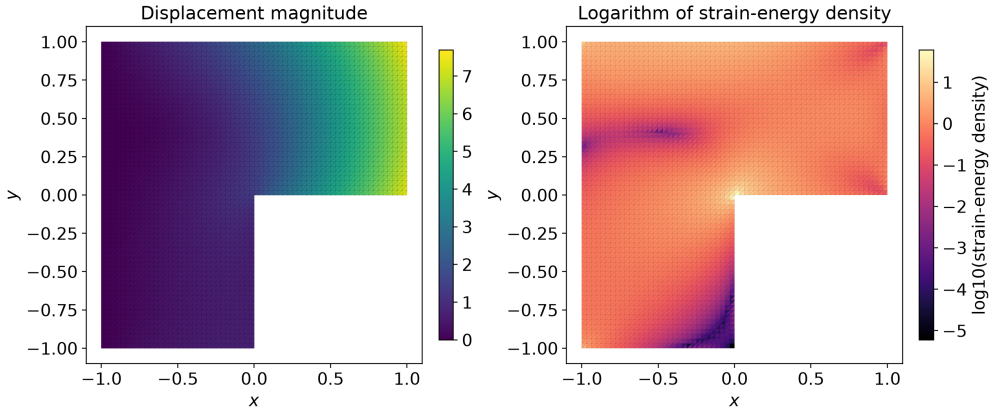



Remove the lower-right quadrant from $(-1, 1)^2$.

- Prescribe $\boldsymbol{u} = \boldsymbol{0}$ on $x = -1$.
- Apply $\boldsymbol{t} = (0, -1)$ on the segment $x = 1$, $0 \leq y \leq 1$.
- Use zero traction on the remaining boundary.

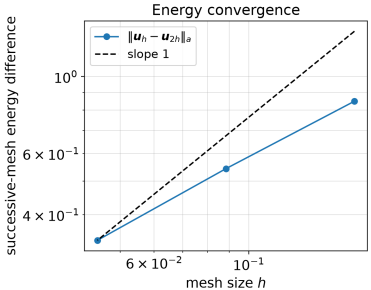
We do not know an exact displacement for this problem. It is therefore not used to calculate an exact-error convergence rate.

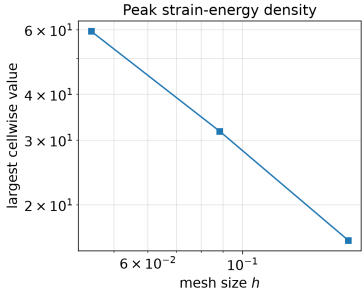
# The corner produces a localized field




The displacement varies across the body, while the strain-energy density is strongly concentrated near the origin. The energy density is shown on a logarithmic color scale.

# What changes under uniform refinement?




For these nested meshes, define

$$d_h = \|u_h - u_{2h}\|_{a_{el}} .$$

Galerkin orthogonality gives

$$d_h^2 = \ell(u_h) - \ell(u_{2h}).$$

The earlier rate formula, $r_h = \log(d_h/d_{h/2})/\log 2$, gives approximately 0.65 and 0.69, below the smooth-solution rate 1.

The largest cellwise strain-energy density increases under refinement, showing stronger localization near the corner.

# The corner limits the observed convergence rate

For linear elements, the estimate derived earlier gives

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{a_{\mathrm{el}}} \leq Ch\,|\boldsymbol{u}|_{H^2(\Omega)}$$

provided the exact displacement belongs to $[H^2(\Omega)]^2$.

The successive-mesh energy differences, rather than exact errors, decrease and provide numerical evidence of global energy convergence. Their observed rates, approximately 0.65 and 0.69, are nevertheless below the smooth-solution rate 1. This is consistent with the reentrant corner reducing the regularity of the exact displacement.

At the same time, the largest cellwise strain-energy density increases near the corner. Global energy convergence therefore does not prevent a local strain and stress concentration. Chapter 8 uses residual indicators to identify where local refinement is needed.

# Summary

- Céa's lemma and interpolation give energy-, $H^1$-seminorm, and $H^1$-norm estimates proportional to $h^{s-1}$.
- If the dual problem is $H^2$-regular, Galerkin orthogonality introduces the interpolation error $z - \mathcal{I}_h z$ and supplies one additional power of $h$ in the $L^2$ estimate.
- The smooth $P_1$ elasticity example exhibits rates 2 in $L^2$ and 1 in the $H^1$ seminorm and energy norm.
- For the L-shaped example, successive-mesh energy differences converge at rates near 0.7, below the smooth $P_1$ energy rate 1.
- The reentrant corner reduces regularity and produces a local strain-energy concentration, motivating local refinement.